

Hongjie Fang

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Education

Shanghai Jiao Tong University <i>Ph.D. Student, Wu Wenjun Honorable Class, Computer Science and Engineering</i>	2022/09 – present Shanghai, China
Shanghai Jiao Tong University <i>Bachelor of Engineering, major in Computer Science and Engineering, Bachelor of Economics, minor in Finance</i> • GPA 4.03 / 4.3, Ranking: 2 / 149.	2018/09 – 2022/06 Shanghai, China

Awards

Best Paper Award , ICRA 2024	2024
Best Paper Award , IROS 2023 Manipulation and Grasping Workshop	2023
Shanghai Outstanding Graduates	2022
Shanghai Scholarship	2020
National Olympics in Informatics (NOI) <i>Bronze Medal</i>	2017
National Olympics in Informatics in Provinces (NOIP) <i>First Prize</i>	2015 & 2016

Publications & Preprints

Total Citations: 3402

1. Force Policy: Learning Hybrid Force-Position Control Policy under Interaction Frame for Contact-Rich Manipulation
Hongjie Fang*, Shirun Tang*, Mingyu Mei*, Haoxiang Qin, Zihao He, Jingjing Chen, Ying Feng, Chenxi Wang, Wanxi Liu, Zaixing He, Cewu Lu, Shiquan Wang
Robotics: Science and Systems (RSS), 2026.

2. AirExo-2: Scaling up Generalizable Robotic Imitation Learning with Low-Cost Exoskeletons
Hongjie Fang*, Chenxi Wang*, Yiming Wang*, Jingjing Chen*, Shangning Xia, Jun Lv, Zihao He, Xiyan Yi, Yunhan Guo, Xinyu Zhan, Lixin Yang, Weiming Wang, Cewu Lu, Hao-Shu Fang
Conference on Robot Learning (CoRL), 2025, **(Oral Presentation)**.

3. AirExo: Low-Cost Exoskeletons for Learning Whole-Arm Manipulation in the Wild
Hongjie Fang*, Hao-Shu Fang*, Yiming Wang*, Jieji Ren, Jingjing Chen, Ruo Zhang, Weiming Wang, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2024.

4. TransCG: A Large-Scale Real-World Dataset for Transparent Object Depth Completion and A Grasping Baseline
Hongjie Fang, Hao-Shu Fang, Sheng Xu, Cewu Lu
IEEE Robotics and Automation Letters (RA-L), 2022; *Presented at ICRA 2023*.

5. History-Aware Visuomotor Policy Learning via Point Tracking
Jingjing Chen*, **Hongjie Fang***, Chenxi Wang, Shiquan Wang, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2026.

6. FoAR: Force-Aware Reactive Policy for Contact-Rich Robotic Manipulation
Zihao He*, **Hongjie Fang***, Jingjing Chen, Hao-Shu Fang, Cewu Lu
IEEE Robotics and Automation Letters (RA-L), 2025; *Presented at IROS 2025*.

7. Learning Dexterous Manipulation with Quantized Hand State
Ying Feng*, **Hongjie Fang***, Yinong He*, Jingjing Chen, Chenxi Wang, Zihao He, Ruonan Liu, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2026.

8. RH20T: A Comprehensive Robotic Dataset for Learning Diverse Skills in One-Shot
Hao-Shu Fang, **Hongjie Fang**, Zhenyu Tang, Jirong Liu, Chenxi Wang, Junbo Wang, Haoyi Zhu, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2024

9. RISE: 3D Perception Makes Real-World Robot Imitation Simple and Effective
Chenxi Wang, **Hongjie Fang**, Hao-Shu Fang, Cewu Lu
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.

10. CAGE: Causal Attention Enables Data-Efficient Generalizable Robotic Manipulation
Shangning Xia, **Hongjie Fang**, Hao-Shu Fang, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2025.
11. Towards Effective Utilization of Mixed-Quality Demonstrations in Robotic Manipulation via Segment-Level Selection and Optimization
Jingjing Chen, **Hongjie Fang**, Hao-Shu Fang, Cewu Lu
IEEE International Conference on Robotics and Automation (ICRA), 2025.
12. AnyGrasp: Robust and Efficient Grasp Perception in Spatial and Temporal Domains
Hao-Shu Fang, Chenxi Wang, **Hongjie Fang**, Minghao Gou, Jirong Liu, Hengxu Yan, Wenhai Liu, Yichen Xie, Cewu Lu
IEEE Transaction on Robotics (T-RO), 2023; Presented at ICRA 2024, (**Best Paper Award on IROS 2023 Workshop**).
13. Motion Before Action: Diffusing Object Motion as Manipulation Condition
Yue Su*, Xinyu Zhan*, **Hongjie Fang**, Yong-Lu Li, Cewu Lu, Lixin Yang
IEEE Robotics and Automation Letters (RA-L), 2025; Presented at ICRA 2026.
14. Dense Policy: Bidirectional Autoregressive Learning of Actions
Yue Su*, Xinyu Zhan*, **Hongjie Fang**, Han Xue, Hao-Shu Fang, Yong-Lu Li, Cewu Lu, Lixin Yang
IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
15. Knowledge-Driven Imitation Learning: Enabling Generalization Across Diverse Conditions
Zhuochen Miao*, Jun Lv*, **Hongjie Fang**, Yang Jin, Cewu Lu
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.
16. Open X-Embodiment: Robotic Learning Datasets and RT-X Models
Open X-Embodiment Collaboration, 147 authors
IEEE International Conference on Robotics and Automation (ICRA), 2024, (**Best Paper Award**).
17. Flexible Handover with Real-Time Robust Dynamic Grasp Trajectory Generation
Gu Zhang, Hao-Shu Fang, **Hongjie Fang**, Cewu Lu
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.
18. AnyDexGrasp: Learning General Dexterous Grasping for Any Hands with Human-level Learning Efficiency
Hao-Shu Fang, Hengxu Yan, Zhenyu Tang, **Hongjie Fang**, Chenxi Wang, Cewu Lu
arXiv preprint, 2025.
19. LIDEA: Human-to-Robot Imitation Learning via Implicit Feature Distillation and Explicit Geometry Alignment
Yifu Xu*, Bokai Lin*, Xinyu Zhan, **Hongjie Fang**, Yong-Lu Li, Cewu Lu, Lixin Yang
arXiv preprint, 2026.
20. SIME: Enhancing Policy Self-Improvement with Modal-Level Exploration
Yang Jin*, Jun Lv*, Wenye Yu, **Hongjie Fang**, Yong-Lu Li, Cewu Lu
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.
21. Graspness Discovery in Clutters for Fast and Accurate Grasp Detection
Chenxi Wang*, Hao-Shu Fang*, Minghao Gou, **Hongjie Fang**, Jin Gao, Cewu Lu
IEEE/CVF International Conference on Computer Vision (ICCV), 2021.
22. Target-Referenced Reactive Grasping for Dynamic Objects
Jirong Liu, Ruo Zhang, Hao-Shu Fang, Minghao Gou, **Hongjie Fang**, Chenxi Wang, Sheng Xu, Hengxu Yan, Cewu Lu
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.

Academic Services

Reviewer for journal *RA-L*, *T-CYB*, *T-MECH*.

Reviewer for conferences including *ICRA*, *IROS*, *CoRL*, *NeurIPS*, *ICLR*, *CVPR*, etc.

Invited Talks

- 03/2024, **Echo AI @ USyd**, *Towards Efficient Robot Imitation Learning from Human Demonstrations*.
- 11/2024, **Zhixingxing**, *Towards Efficient Robot Imitation Learning from Human Demonstrations*.

- 02/2025, **Yang Gao Group @ THU**, *Towards Generalizable Imitation Learning from Human Demonstrations*.
- 08/2025, **3D CVer**, *Towards Generalizable Imitation Learning from Human Demonstrations*.
- 08/2025, **Sharpa**, *Towards Generalizable Imitation via Scalable Data and Robust Policy*.
- 09/2025, **Galbot**, *Building Capable and Generalizable Policies for Robotic Manipulation*.
- 09/2025, **Shenlanxueyuan**, *Towards Generalizable Imitation via Scalable Data and Robust Policy*.
- 10/2025, **The Robot Dexterity Lab @ Duke**, *AirExo-2: Scaling up Generalizable Robotic Imitation Learning with Low-Cost Exoskeletons*.

Teaching

Teaching Assistant, <i>Algorithm and Complexity</i>	<i>Spring, 2022</i>
Teaching Assistant, <i>C++ Programming Language (Honor)</i>	<i>Fall, 2020</i>
Teaching Assistant, <i>Data Structure (Honor)</i>	<i>Spring, 2019</i>
Teaching Assistant, <i>Linear Algebra (Honor)</i>	<i>Fall 2021 & Fall, 2022</i>
Teaching Assistant, <i>Mathematical Analysis (Honor)</i>	<i>Fall, 2020 & Fall, 2021</i>